Number Systems

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- A number system (or numeral system) defines how a number can be represented using distinct symbols.
- A number can be represented differently in different systems.
- In this system, a number represented as:

$$\pm(S_{k-1} \dots S_2 S_1 S_0 . S_{-1} S_{-2} \dots S_{-l})_b$$

has the value of :

$$v = \pm\left(S_{k-1} \times b^{k-1} + \cdots + S_1 \times b^1 + S_0 \times b^0 + S_{-1} \times b^{-1} + S_{-2} b^{-2} + \cdots + S_{-l} \times b^{-l}\right)$$

$$(2\,3\,5\,7.\,8\,4)_{10}$$

$10^2\ \ 10^0\ \ 10^{-2}$

$10^3\ \ 10^1\ \ 10^{-1}$

$$=2*10^3+\ 3*10^2+\ 5*10^1+\ 7*10^0+\ 8*10^{-1}\ +\ 4*10^{-2}$$

- The decimal system (base 10)

  $S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

- The binary system (base 2)

  $S = \{0, 1\}$

- The hexadecimal system (base 16)

  $S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F\}$

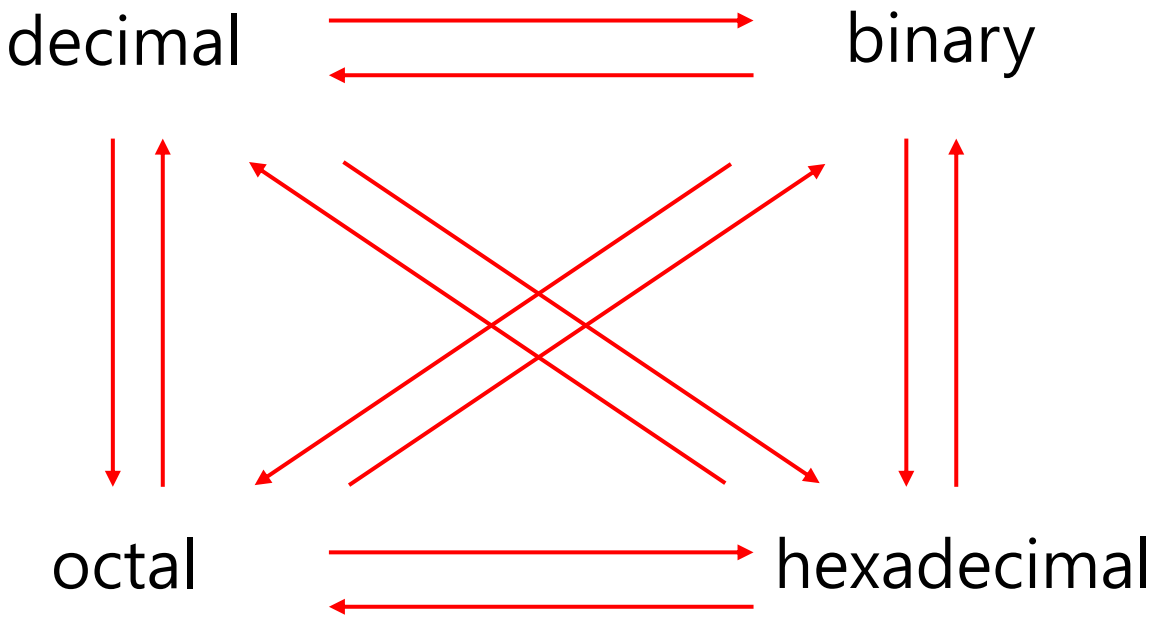
- The octal system (base 8)

  $S = \{0, 1, 2, 3, 4, 5, 6, 7\}$

- The decimal system
  - $(23576)_{10} = 23576$
- The binary system
  - $(10101)_2$, $(10101.0111)_2$
- The hexadecimal system
  - $(235BE)_{16}$, $(235BE.65A)_{16}$
- The octal system (base 8)
  - $(23576)_8$, $(2357.572)_8$

decimal	binary	octal	hexadecimal
00	0000	00	00
01	0001	01	01
02	0010	02	02
03	0011	03	03
04	0100	04	04
05	0101	05	05
06	0110	06	06
07	0111	07	07
08	1000	10	08
09	1001	11	09
10	1010	12	0A
11	1011	13	0B
12	1100	14	0C
13	1101	15	0D
14	1110	16	0E
15	1111	17	0F

Conversion



- Decimal

$$(\underset{10^4}{2}\ \underset{10^3}{3}\ \underset{10^2}{5}\ \underset{10^1}{7}\ \underset{10^0}{6})_{10}$$

$$2*10^4 + 3*10^3 + 5*10^2 + 7*10^1 + 6*10^0$$

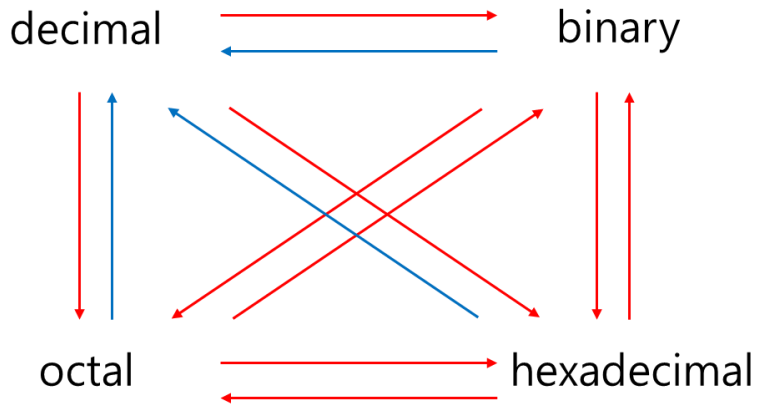
- Binary

$$(\underset{2^4}{1}\ \underset{2^3}{0}\ \underset{2^2}{1}\ \underset{2^1}{0}\ \underset{2^0}{1})_2$$

$$1*2^4 + 0*2^3 + 1*2^2 + 0*2^1 + 1*2^0$$

$$= 16 + 0 + 4 + 0 + 1$$

$$= 21$$



- Octal

$$(2\ 3\ 5\ 7\ 6)_8$$

$8^4\ 8^3\ 8^2\ 8^1\ 8^0$

$2*8^4+ 3*8^3+ 5*8^2+ 7*8^1+ 6*8^0$

$=2*4096+3*512+5*64+7*8+6*1$

$=8192+1536+320+56+6 = 10110$

- Hexadecimal

$$(2\ 3\ 5\ B\ E)_{16}$$

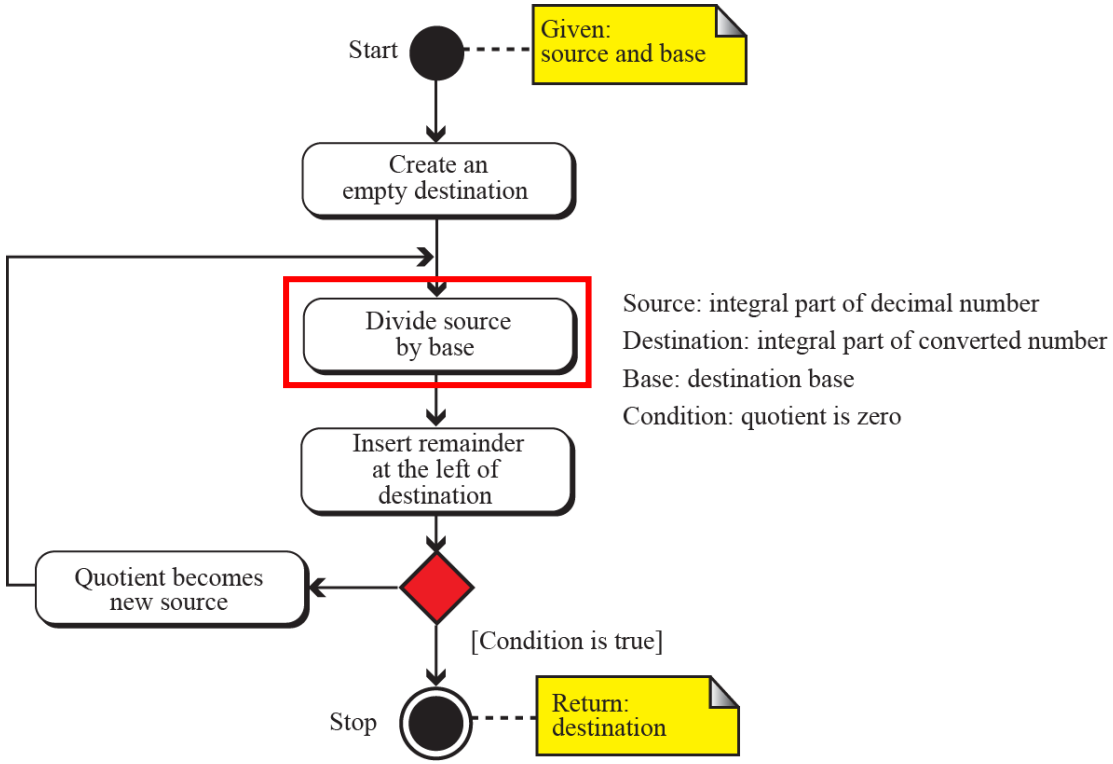
$2*16^4+ 3*16^3+ 5*16^2+ 11*16^1+ 14*16^0$

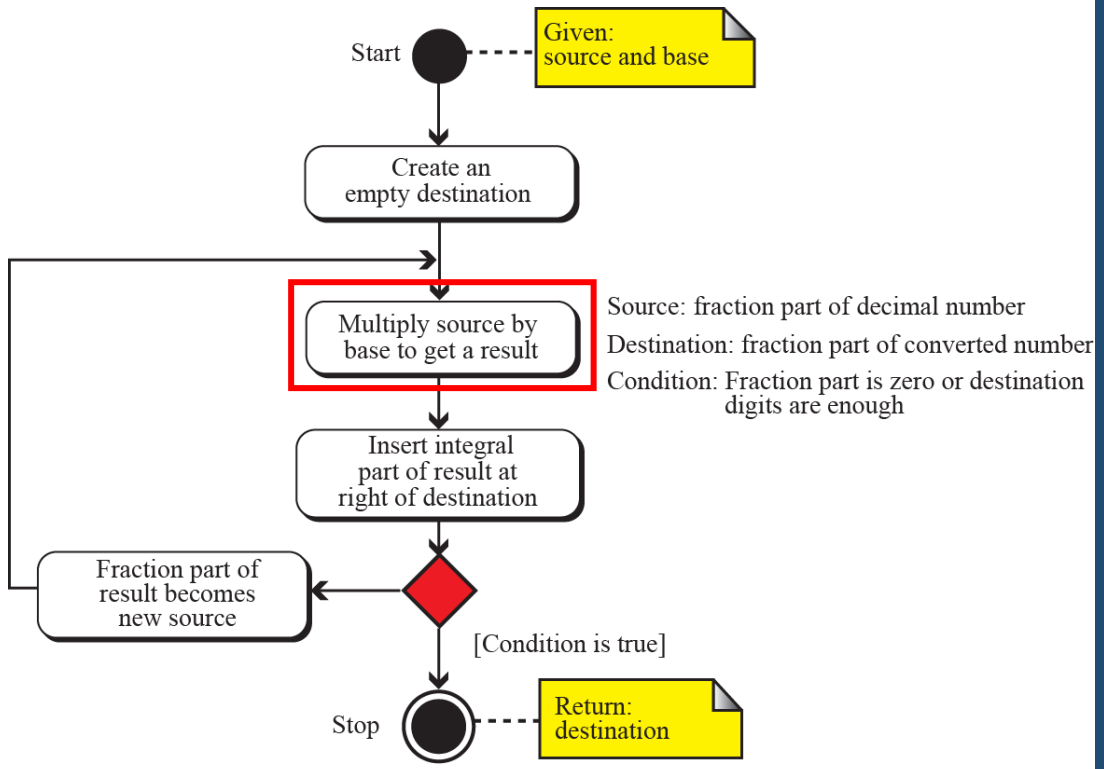
$= 2*65536+3*4096+5*256+11*16+14*1$

$=131072+12288+1280+176+14 =144830$

› Decimal to any number systems

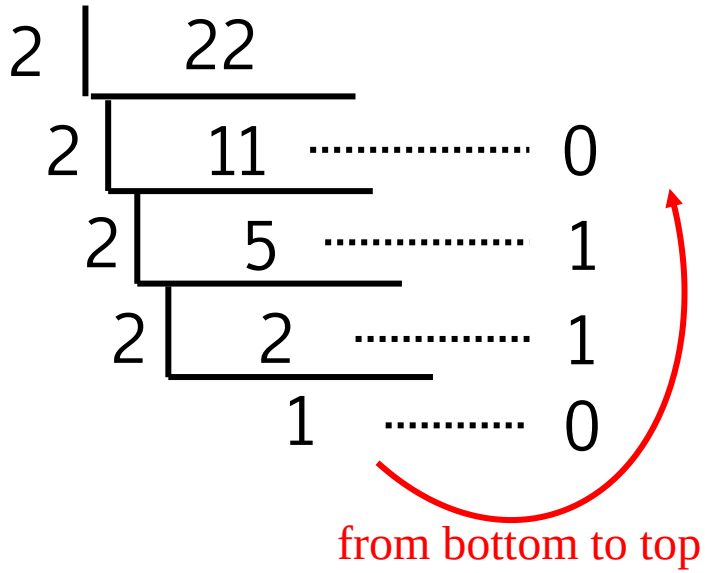
- integral part
- fractional part



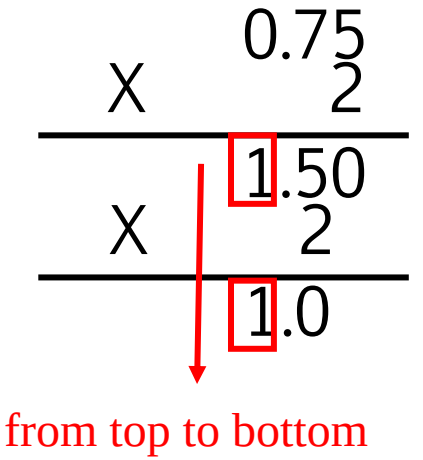


Convert 22.75 in decimal to binary

$(22.75)_{10}=(\ 10110.11\ )_2$

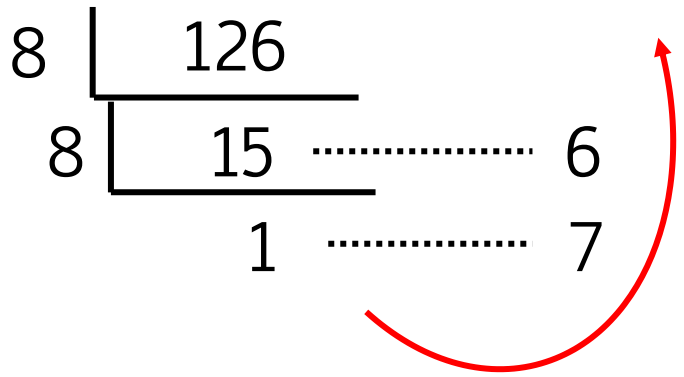


$(22)_{10}=(10110)_2$



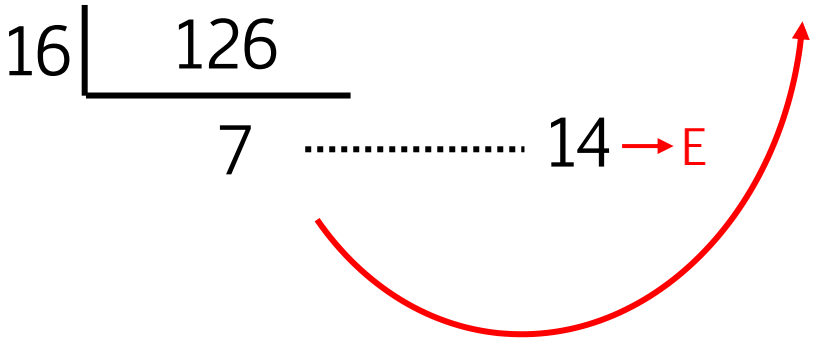
$(0.75)_{10}=(0.11)_2$

Convert 126 in decimal to octal

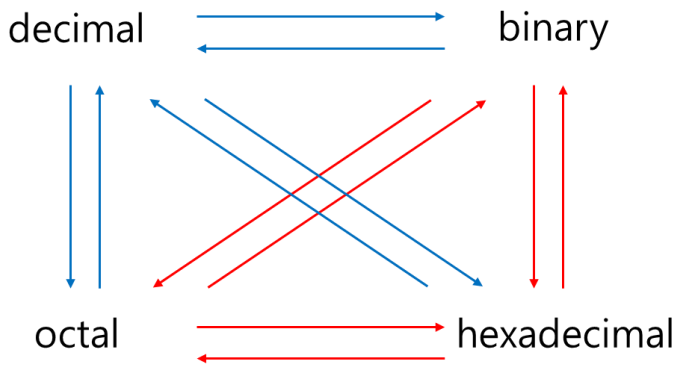


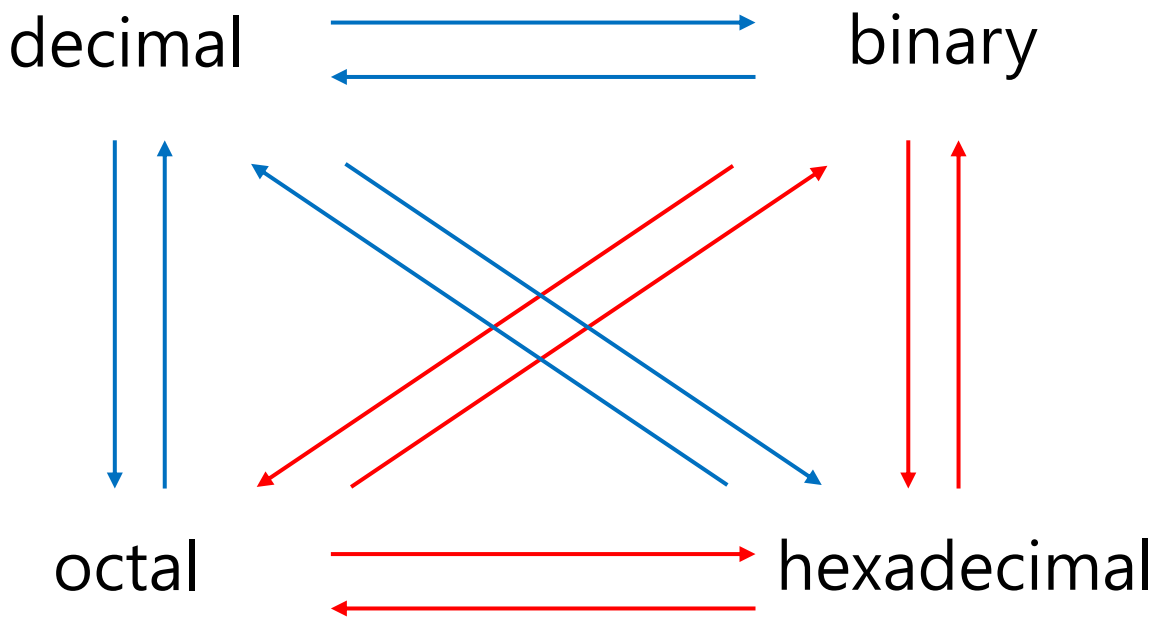
$(126)_{10}=(176)_{8}$

Convert 126 in decimal to hexadecimal



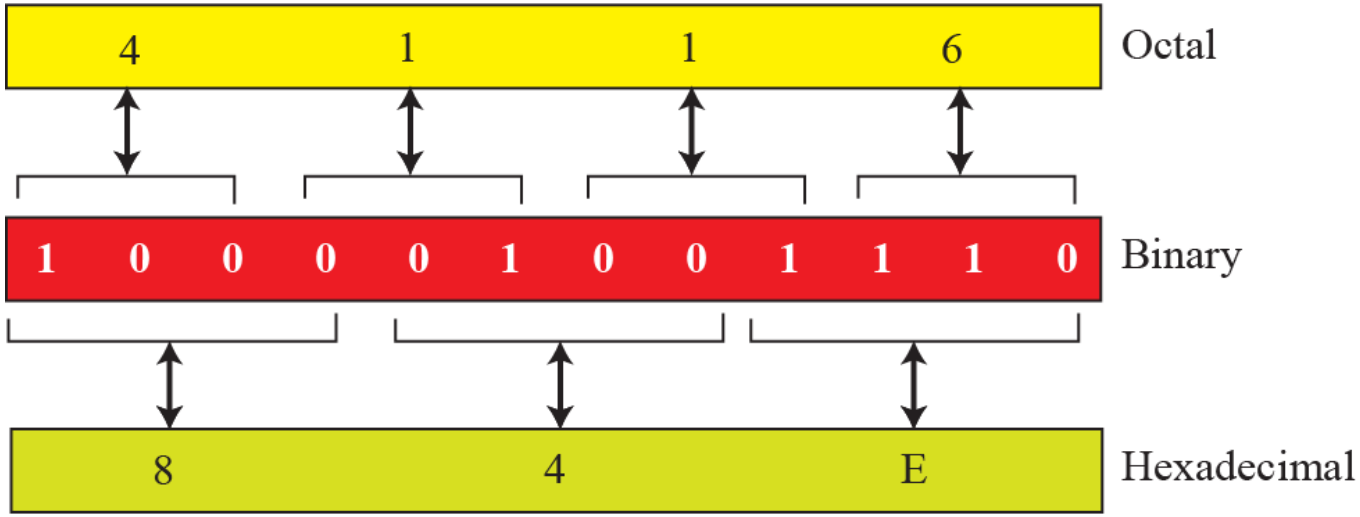
$(126)_{10}=(7E)_{16}$



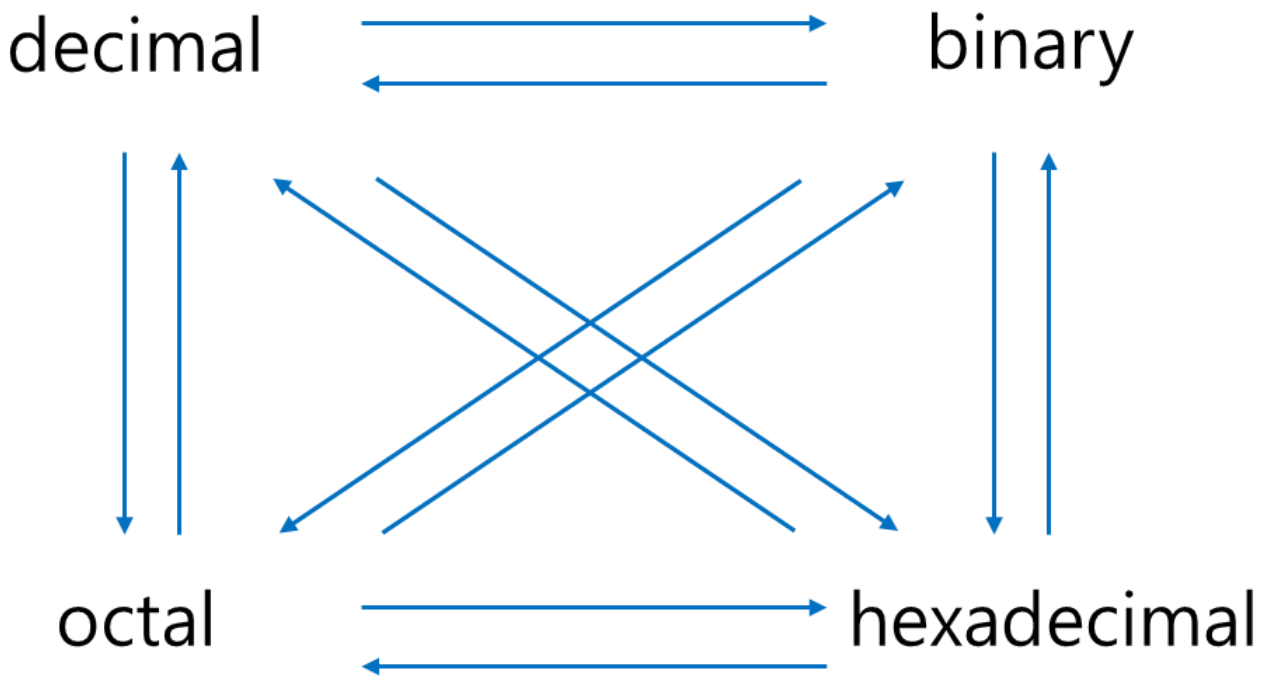


binary	octal	hexadecimal
2	8	16
2	2^3	2^4

- Binary to octal and octal to binary conversion
- Binary to hexadecimal and hexadecimal to binary conversion



- Octal to hexadecimal and hexadecimal to octal conversion


decimal
binary
octal
hexadecimal

Representing Integers

- Unsigned integers can be represented in base two
- Signed integers : numbers that can be positive or negative
  - Signed magnitude
  - One's complement
  - Two's complement
    - the most popular representation
- Floating-point
  - representing integer with fractional

$(1\,0\,1\,1\,0\,0\,1\,1\,0\,1\,1\,1\,0\,1\,0)_2$

MSB
most significant bit

LSB
least significant bit

› Signed magnitude

– $-(2^{(n-1)}-1) \sim 2^{(n-1)}-1$

000	+0
001	+1
010	+2
011	+3
100	-0
101	-1
110	-2
111	-3

› One's complement

– $-(2^{(n-1)}-1) \sim 2^{(n-1)}-1$

000	+0
001	+1
010	+2
011	+3
100 011	-3
101 010	-2
110 001	-1
111 000	-0

- Two's complement
  - $-(2^{(n-1)})\sim 2^{(n-1)}-1$

000	+0
001	+1
010	+2
011	+3
100 011+1	-4
101 010+1	-3
110 001+1	-2
111 000+1	-1

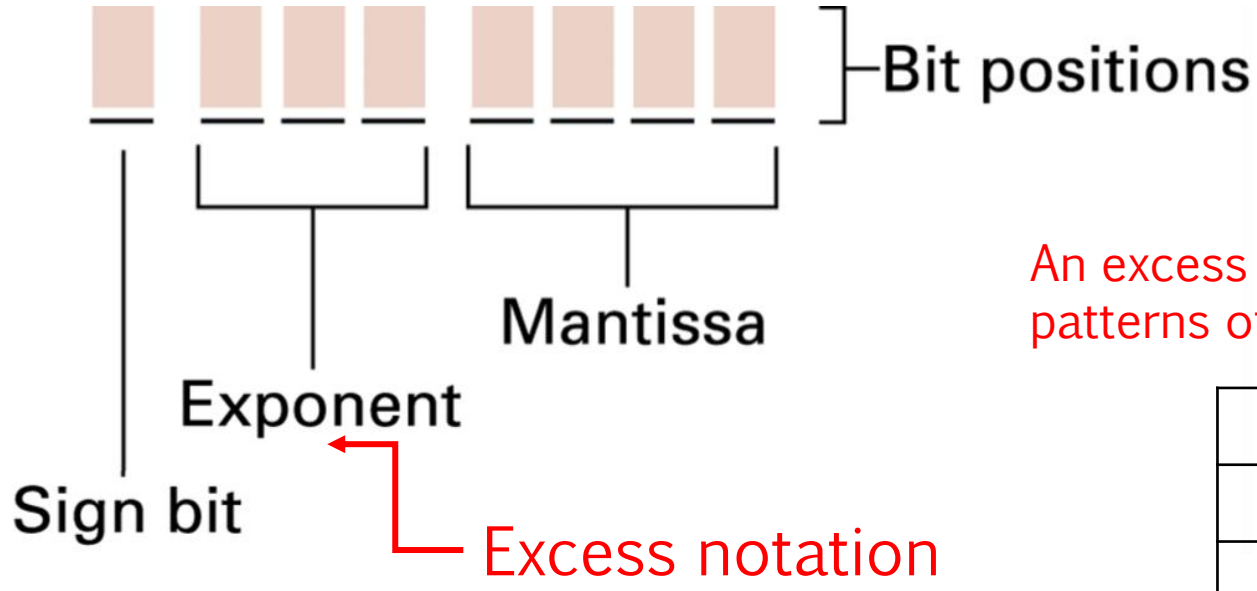
0000	+0
0001	+1
0010	+2
0011	+3
0100	+4
0101	+5
0110	+6
0111	+7
1000	-8
1001	-7
1010	-6
1011	-5
1100	-4
1101	-3
1110	-2
1111	-1

Number	Binary	Signed Magnitude	1's complement	2's complement
0	000	+0	+0	+0
1	001	+1	+1	+1
2	010	+2	+2	+2
3	011	+3	+3	+3
4	100	-0	-3	-4
5	101	-1	-2	-3
6	110	-2	-1	-2
7	111	-3	-0	-1

Example

- Represent the value -22 in signed magnitude, one's complement and two's complement

Floating-point notation

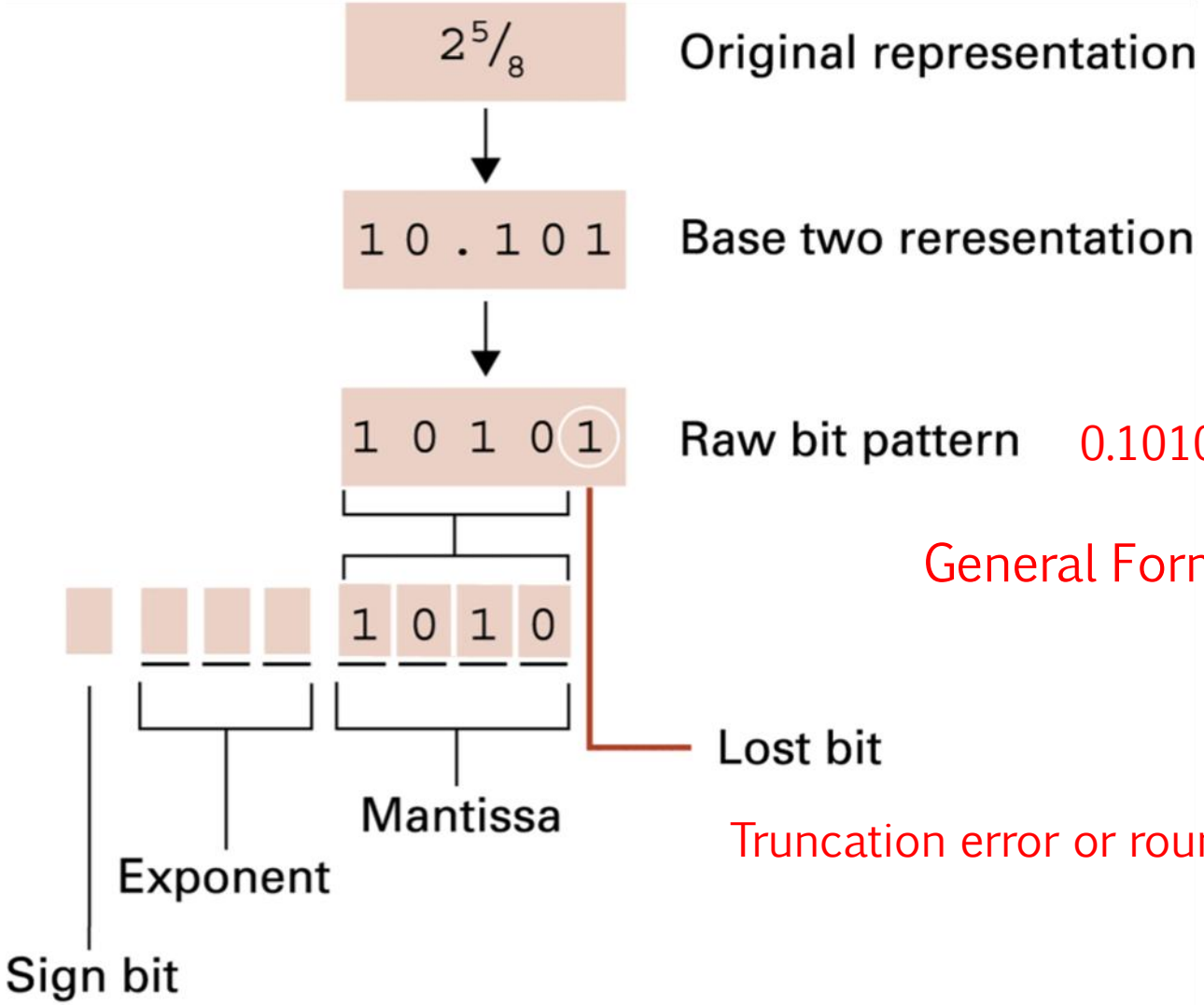


An excess notation system using bit patterns of length three

111	3
110	2
101	1
100	0
011	-1
010	-2
001	-3
000	-4

General Form: $\pm m \times R^e$

Coding the value $2\frac{5}{8}$



General Form: $\pm m \times R^e$

Truncation error or round-off error

IEEE 754 Floating Point

› Single-precision floating-point format (32 bits)



– Exponent bias is 127

› Double-precision floating-point format (64 bits)



– Exponent bias is 1023

Coding the value 12.625 in IEEE 754 single-precision floating-point format

- 12.625
  $=(1100.101)_2$
  $=1.100101*2^3$

  - Sign: 0
  - Exponent: 3+127=130=$(10000010)_2$
  - Mantissa: 10010100000000000000000

- 01000001010010100000000000000000

Sign Exponent Mantissa

- Q&A…